

ARUNKUMAR K R

✉ arun8778jul@gmail.com | ☎ +91 87787 65734 | 📁 Portfolio | in LinkedIn
🐙 GitHub | 💻 Leetcode | 🏠 Hackerrank

SUMMARY

Computer Science Engineering undergraduate specializing in AI-powered applications and backend development. Skilled in building scalable APIs, intelligent automation systems, and practical AI solutions using Python, FastAPI, Node.js and LLM-based technologies. Experienced in developing real-world software products with a focus on performance and system efficiency. Collaborative team player with research experience in Computer Vision and RHCSA certification.

EDUCATION

KPR Institute of Engineering and Technology , Coimbatore	Sept 2023 – Apr 2027
B.E. Computer Science and Engineering	CGPA: 8.77 / 10
Kongu Vellalar Matriculation Higher Secondary School , Perundurai	Jun 2022 – Mar 2023
Higher Secondary Education (Class XII)	94.6%

SKILLS

- **Languages:** Python, Java, JavaScript, C, SQL
- **Backend Development:** Node.js, Express.js, FastAPI, Django
- **AI & Machine Learning:** LangChain, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, ChromaDB
- **Databases & Infrastructure:** PostgreSQL, MongoDB, Redis, Docker, Linux, Git
- **Soft skills:** Problem Solving, Team Collaboration, Communication, Leadership, Adaptability

EXPERIENCE

Reccsar Pvt Ltd	Jan 2025 – Feb 2025
<i>Backend Developer Intern</i>	
• Built and shipped 5+ production features using React.js, Node.js, and Express.js, improving workflows for 100+ active users. • Connected 10+ REST APIs to a PostgreSQL backend, reducing manual data entry effort by roughly 30%.	

PROJECTS

Prism AI – Content Generation Platform	Live Demo GitHub
<i>Python, FastAPI, Groq LLM, Hugging Face FLUX.1</i>	
• Developed a multi-modal AI platform that generates SEO blog posts, video scripts, and images using Groq LLM and Hugging Face FLUX via async APIs. • Unified the frontend and all AI services under a single FastAPI backend, simplifying infrastructure while supporting concurrent AI generation tasks.	
RouteX – Intelligent PDF Routing System	Live Demo GitHub
<i>Python, PyMuPDF, Tesseract OCR, NLP</i>	
• Built a document processing pipeline using OCR and NLP to classify both scanned and digital PDFs with 95% accuracy. • Added automatic tagging via NLP summarization and TF-IDF keyword extraction, significantly reducing manual categorization effort.	
ProManage – Project Management API	Live Demo GitHub
<i>Node.js, TypeScript, PostgreSQL, Redis, JWT</i>	
• Built a scalable REST API with Role-Based Access Control and JWT authentication, supporting distinct permission levels for admins, managers, and members. • Cut API response time by 45% using Redis caching and Node.js Streams, maintaining low latency under 500+ concurrent requests.	

RESEARCH & ACHIEVEMENTS

- **Research Publication:** "Real-Time Surface Defect Detection in Metals using YOLOv8" – presented and published at the **3rd ICEAMST International Conference**; focused on industrial CV applications with deep learning.
- **RedHat Certified System Administrator (RHCSA)** – Industry-recognized Linux system certification.
- **Competitive Programming:** Solved **500+ DSA problems** on LeetCode and HackerRank, with verified certifications in Problem Solving (Intermediate), Python, Java, and SQL.
- **Leadership:** Executive Member at Dept. Association – coordinated technical events and workshops for 100+ students.
- **Hobbies:** Playing cricket, Music, Swimming